

Four thesis directions

Each one extends the published DhakaSim and RoadBird work, carries a real machine learning component, and connects back to your earlier study of lane diversion and intersection blocking at a four-legged junction. They are ordered by how much risk they carry, lowest first.

The expected findings in each proposal are hypotheses, not promises. Two of them are set up so that a negative result is still publishable, which matters when you have a fixed submission date.

A learned surrogate of the simulator, and what it reveals about diversion

Objective

Build a graph neural network that predicts per-link speed, waiting time and flow from a network, a demand pattern and a diversion configuration, without running the simulator. Then use it to answer which links, under which demand, most determine how the whole network performs.

Methodology

1. Make strip width, diversion and intersection blocking per-link properties rather than one global setting, so a configuration becomes a vector over the network rather than a single switch.
2. Instrument the simulator to log link state at fixed intervals instead of only writing an average at the end of a run.
3. Generate training data by sweeping configurations, demand levels and topologies, using the parallel harness already in the repository.
4. Train a message-passing network over the road graph, with links as nodes and junctions as edges. Compare against ridge regression and gradient boosting on hand-built features.
5. Apply permutation importance and a graph-specific attribution method to find which links dominate the prediction, and cluster links by their response profile.

Expected findings

- A surrogate accurate enough to replace the simulator for screening, with error reported using the same measures the 2024 paper uses in its Table IV, so the numbers are directly comparable to published ones.
- A small subset of links that account for most of the network's performance. This is the practically useful output, because it says where intervention is worth paying for.
- A typology of link behaviour: links that saturate early, links that only matter once diversion is applied, and links that never bind.

Novelty

No paper in this line builds any model of the simulator. The 2025 scheduling paper searches with NSGA-II, which explores a space without learning its shape. A surrogate is a different object and enables work none of these papers could attempt.

Lowest risk of the four. The data is generated, so nothing depends on fieldwork or on an external release. If the surrogate turns out to be inaccurate, that is itself a reportable finding about how much of this traffic is predictable from network structure.

Forecasting congestion and diverting before it forms

Objective

Predict where congestion will appear several minutes ahead from the network's current state, and use that forecast to trigger diversion in advance rather than in response. Your earlier work applied diversion as a fixed condition. This makes it a decision taken at a moment in time.

Methodology

1. Log per-link time series from the simulator, giving speed, occupancy and queue state at regular intervals.
2. Train a spatio-temporal graph model, combining message passing over the road graph with a recurrent or temporal convolution component, to forecast link state at several horizons.
3. Define a trigger policy on top of the forecast: when predicted congestion on a link exceeds a threshold, divert the affected route.
4. Compare three regimes in the simulator: no diversion, reactive diversion once congestion has formed, and predictive diversion driven by the forecast.
5. Sweep the forecast horizon to find how far ahead the prediction has to reach before acting early is better than acting late.

Expected findings

- A horizon beyond which forecasting stops helping, because the prediction degrades faster than the extra warning time is worth. Locating that point is the headline result.
- Predictive diversion outperforming reactive diversion on waiting time, with the gap widening as demand rises.
- Evidence about how far congestion propagates between adjacent links in non-lane traffic, which is not documented anywhere in the existing papers.

Novelty

Spatio-temporal traffic forecasting is a large field, but it is almost entirely lane-based and trained on loop detector or probe data from developed cities. Doing it for non-lane heterogeneous traffic, with side friction modelled, and then closing the loop by acting on the forecast, is not something these papers or the wider forecasting literature covers.

Moderate risk. The forecasting model is standard enough that it will train, and the control loop is simple. The main uncertainty is whether the simulator produces congestion dynamics rich enough to be worth forecasting at all, which you can check cheaply before committing.

Does faster traffic mean more dangerous traffic?

Objective

Measure the safety consequences of the interventions the existing papers recommend for throughput, and test whether the two objectives conflict. Every paper in this line measures speed, waiting time and flow. None measures safety, even though the simulator already counts vehicle and pedestrian encounters and carries a time-to-collision threshold.

Methodology

1. Extract surrogate safety measures from the simulator, using encounter counts and time-to-collision below threshold, separated by whether a pedestrian is involved.
2. Build a classifier that predicts conflict risk per link from traffic state and configuration. The positive class will be rare, so this is an imbalanced learning problem and should be treated as one, with precision-recall reporting rather than accuracy.
3. Re-run the interventions the published papers recommend, including lane conversion, diversion and the optimised signal schedules, and measure safety alongside throughput.
4. Map the trade-off surface between average speed and conflict rate, and identify configurations that improve one without harming the other.
5. Use attribution over the classifier to identify which conditions drive pedestrian conflict specifically.

Expected findings

- A speed and safety trade-off, where configurations that raise average speed also raise conflict rate. If that holds, several recommendations in the existing literature look different once safety is priced in.
- Pedestrian conflict concentrated in a small number of conditions rather than spread evenly, which would make it addressable.
- Some configurations that improve both. Those are the valuable ones and the reason to map the surface rather than assume the trade-off is strict.

Novelty

This is the clearest gap in the six papers. The 2025 side friction paper establishes that road-crossing pedestrians are the largest single source of congestion in Dhaka, which sets the question up directly, and then measures only congestion.

Moderate risk, concentrated in one place. The simulator's conflict counts are a surrogate for real safety, not a measurement of it, and you would need to argue that case carefully rather than assert it. Check early how the encounter model actually behaves, because the whole direction rests on it.

DIRECTION 4

Automatic calibration, and whether a model of one city transfers to another

Objective

Replace hand calibration with a systematic procedure that fits simulator parameters to observed data, then test whether parameters fitted for one city carry over to another. The published work calibrates by hand in every paper and reports the accuracy achieved, but nobody has made the procedure repeatable.

Methodology

1. Treat calibration as an inverse problem: given observed link speeds or travel times, recover the parameter set most likely to have produced them.
2. Use Bayesian optimisation over the parameter space, with the simulator as the expensive objective, and the error measures already implemented in `experiments/stats.py` as the loss.
3. Train a surrogate as in Direction 1 to make the inner loop affordable, since each simulator evaluation costs several minutes.
4. Quantify which parameters the data actually identifies. Some will be pinned tightly by the observations and others will be unconstrained, and knowing which is which is a result in itself.
5. Fit on one city's network and evaluate on another, using the Dhaka, Miami and Riyadh topologies, to measure how far a calibration transfers.

Expected findings

- Calibration accuracy matching or exceeding the hand-tuned figures in the published papers, but produced by a procedure someone else can rerun.
- A subset of parameters that the observations cannot identify at all. This would be an uncomfortable but valuable finding, because it means published hand-calibrated values for those parameters are not evidence of anything.
- Partial transfer between cities, with the parameters governing driver behaviour transferring less well than those governing geometry.

Novelty

Methodological rather than applied, and it addresses the standard criticism of simulator-based transport papers, which is that the calibration is bespoke and unrepeatable. It also strengthens every other result in the group's line of work, which makes it easy to justify.

Highest risk, for one reason. Fitting to observed data requires observed data. If you cannot obtain real travel times, you can still run the whole thing as recovery of known parameters from synthetic data, which tests the method but weakens the applied claim. Decide this before you commit.

Choosing between them

Direction	Machine learning content	Needs field data	Compute load	Main risk
1. Surrogate and diversion	Graph networks, attribution, clustering	No	High, front-loaded	Low
2. Forecast and pre-empt	Spatio-temporal graph models	No	Moderate	Dynamics may be too simple to forecast
3. Safety trade-off	Imbalanced classification, attribution	No	Moderate	Conflict counts are a surrogate for safety
4. Calibration and transfer	Bayesian optimisation, sensitivity analysis	Yes, for the strong version	High	Data availability

What I would pick

Direction 1 as the spine, with Direction 3 as the second half. The surrogate gives you the machine learning contribution your programme requires and makes everything after it cheap to evaluate. The safety analysis then gives you a finding that people outside this research group will care about, and it uses the same generated data, so the marginal cost is small.

That pairing also survives failure gracefully. If the safety half does not produce a clean result, the surrogate and the link importance analysis are already a complete thesis on their own.

Direction 2 is the one to choose if you would rather work on control than on analysis, and it has the most natural continuity with what you did before, since it turns your fixed diversion into a timed decision.

Direction 4 is the strongest methodological contribution and the one most likely to be cited by the group itself, but do not start it until you know whether you can get observed data.

How your earlier work fits in

All four can use the four-legged intersection as a validation case. In Direction 1, the attribution method should identify the links you already know matter. In Direction 2, the predictive controller should recover the diversion you applied by hand, and ideally apply it sooner. In Direction 3, that intersection gives you a known configuration to measure safety on first. Starting there gives you an early result and shows examiners the continuity between your previous work and the thesis.

One thing to do before choosing

Run a sweep with per-link logging enabled on the Dhaka network and look at the time series. Three of these four directions depend on the simulator producing congestion dynamics with enough structure to model, and an afternoon spent checking that will tell you more than any amount of further reading. If the link states turn out to be nearly constant once the network fills, Direction 2 is not viable and Direction 1 becomes a static regression problem rather than a temporal one.

Prepared from the seven PDFs in DhakaSIM Papers and from measured runs of the Python port in this repository. Expected findings are hypotheses to be tested, not results. Accuracy and impact figures attributed to published papers are theirs.